

Professional Development - Skills Matrix and Action Plan

Professional Development

Self-assessed Skills Matrix across the twelve UoEO skill areas plus six module-specific ML skills, paired with an Action Plan for the rows where I sit at Aware or Trained. Both tables are packaged as a single A4 PDF below.

Skill levels: **Aware** (basic understanding) · **Trained** (applies independently) · **Proficient** (broad and in-depth, low supervision) · **Expert** (subject-matter expert, leads and trains others).

Skills Matrix

Skill area	Level	Evidence
1. Commercial Awareness	Expert	20+ years in IT industry leadership; cyberresiliency and cloud portfolio ownership.
2. Time Management	Proficient	Track 1 submitted two days ahead of deadline; sustained weekly cadence across 12 units.
3. Critical Thinking and Analysis	Proficient	Three-tier decision process used in Track 1; variable-change probe on every tutorial notebook.
4. Communication and Literacy	Proficient	1,000-word executive report; CD1 cross-sector synthesis; commercial sales and pre-sales background.
5. IT and Digital	Expert	IT Architect across enterprise estates; this e-portfolio on Jekyll and GitHub Pages; five-script Python pipeline.
6. Numeracy	Proficient	Regression, clustering and metrics across Units 3-11; MBA quantitative methods.
7. Research	Proficient	Harvard Cite Them Right across ~140 references; literature-anchored cleaning rules; MBA

8. Interpersonal	Expert	dissertation. Group D Coordinator role; pre-sales and sales leadership; multinational distributed teams.
9. Problem-solving	Proficient	20+ years in technical problem-solving roles; found five flaws in my own Individual Presentation work before submission.
10. Ethical Awareness	Proficient	Dual-use CNN argument across Units 9 and 10.
11. Teamwork and Leadership	Expert	Group D Coordinator; engagement-based task allocation (Weick, 1976); 20+ years leading distributed teams.
12. Critical Reflection	Proficient	1,000-word reflective piece (Rolfe et al., 2001); 3 W's reflection across 17 activity cards.
13. Classical ML - Regression	Trained	Linear, multiple and polynomial regression; Track 1 log-price model.
14. Classical ML - Clustering and Similarity	Trained	Jaccard, K-Means with elbow / silhouette / ARI; Track 1 segmentation.
15. Deep Learning - Perceptron and MLP	Trained	Simple perceptron; AND in 6 epochs; 2-3-1 sigmoid XOR by back-propagation.
16. Deep Learning - CNN	Trained	CIFAR-10 trained to 77.82% test accuracy; CNN Explainer 4-case study; individual presentation deliverable (Unit 11).
17. ML Evaluation and Metrics	Trained	Confusion matrix, F1, ROC AUC, MAE / RMSE / R^2 ; parameter-change exercises on AUC and polynomial degree.
18. Explainability	Aware	SHAP-style decomposition on a Matrix-Factorisation prediction; CNN Explainer

pipeline walk-through.

Action Plan (3 to 12 months post-module)

Six development goals targeting the rows where I sit at Aware or Trained, plus selected Proficient rows where I want to push toward Expert.

Skill area	Goal	Action	Timeline
Deep Learning	Trained → Proficient	Self-led CNN project; read Goodfellow, Bengio and Courville (2016) end-to-end.	6 months
Python production ML	Trained → Proficient	Deploy a small ML service with CI/CD on a personal project.	6 months
Explainability	Aware → Trained	Apply SHAP to one deployed model; produce one written case study.	3 months
Ethical Awareness	Proficient → Expert	One position paper on dual-use CNN deployment; monthly EU AI Act tracking.	12 months
Research	Proficient → Expert	One peer-reviewable piece connecting cybersecurity and AI.	12 months
Critical Reflection	Proficient → Expert	Monthly reflective journal post-MSc; three-tier decision process used on one decision note per quarter.	Ongoing

Reviewed and committed by Ariel Mella, 17 May 2026 (v1.1). Next review: end of MSc programme.

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